
Lab 1

Requirements Engineering & UML Use-Case Modelling

Problem Statement #57

Pet Adoption & Shelter Management System

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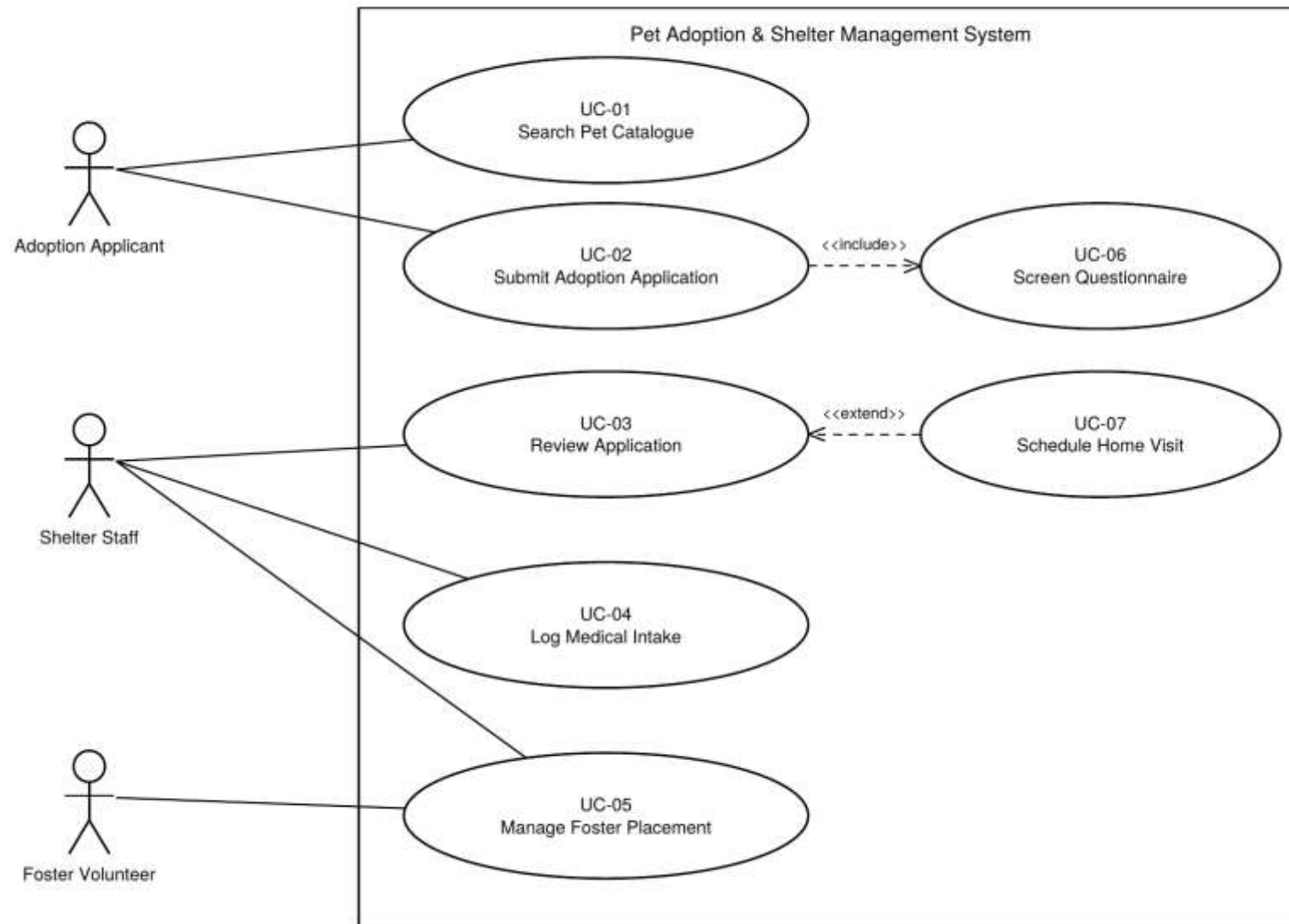
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1. Requirements Table

Problem Statement #57 - Pet Adoption & Shelter Management System

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Shelter Staff to record an animal's vaccination and medical intake details, and shall set the animal's status to "Ready for Adoption" only when all mandatory vaccinations are recorded and no quarantine is active.	High	Pass: A fully vaccinated animal with no active quarantine becomes "Ready for Adoption" and appears in the catalogue. Fail: An animal under quarantine can be marked adopted.	Prevents an unfit animal from being rehomed.
FR-002	Functional	The system shall allow an Adoption Applicant to search the pet catalogue by Species, Age, Breed, Size and "Good with Kids", and to view an animal's detail page.	High	Pass: Species = Dog + Age = 1-3 yrs returns only animals matching both filters. Fail: A non-matching animal appears in the results.	Search is the entry point to the adoption process.
FR-003	Functional	The system shall allow an Adoption Applicant to submit an adoption questionnaire for a selected animal, and shall score it automatically as Auto-Approved, Needs Staff Review or Auto-Rejected.	High	Pass: A score below the rejection threshold is stored as Auto-Rejected with the failing rule recorded. Fail: The form is accepted with a mandatory field blank.	Gives a consistent first screening without staff effort.
FR-004	Functional	The system shall allow Shelter Staff to approve or reject a queued application with a mandatory reason note, and shall email the applicant the outcome within 5 minutes.	High	Pass: Decision saved with reason note, audit entry written, email delivered within 5 minutes. Fail: A decision is saved without a reason note.	Keeps a human decision and an audit trail on record.
FR-005	Functional	The system shall allow Shelter Staff to assign an animal to a Foster Volunteer with a start and end date, and shall list placements ending within the next 7 days on the staff dashboard.	Medium	Pass: A placement ending in 6 days is listed; one ending in 10 days is not. Fail: One animal has two active placements at the same time.	Tracks foster timelines so animals are not lost track of.
NFR-001	Nonfunctional	The pet search catalogue shall return results for any filter combination in under 200 ms, over HTTPS, with 500 concurrent users.	High	Pass: Load test with 500 users shows 95th percentile response time below 200 ms. Fail: Response time reaches 200 ms or more under the target load.	Slow search loses applicants during adoption drives.
NFR-002	Nonfunctional	The system shall allow only Shelter Staff and Veterinarian roles to view medical records and applicant personal data, shall encrypt that data at rest, and shall log every access attempt.	High	Pass: 20 unauthorised requests all return HTTP 403 and each is written to the access log. Fail: A protected record can be opened without an authorised login.	Applications contain personal details the shelter is accountable for.

2. UML Use Case Diagram



3. Use Case Flow

UC-02 Submit Adoption Application

Use Case ID	UC-02
Use Case Name	Submit Adoption Application
Primary Actor	Adoption Applicant
Secondary Actor	Shelter Staff (receives the queued application)
Related Requirements	FR-002, FR-003, NFR-002
Relationships	Includes UC-06 Screen Questionnaire. UC-03 is extended by UC-07 Schedule Home Visit.
Trigger	Applicant clicks "Apply to Adopt" on an animal's detail page
Frequency	Several times a day, higher during adoption drives

Brief Description

An Adoption Applicant applies to adopt an animal that the shelter has cleared as ready for adoption. The applicant fills in a screening questionnaire, the system scores it against the shelter's rules, and the application is either queued for staff review or rejected automatically.

Preconditions

1. The applicant is registered and logged in.
2. The animal's status is "Ready for Adoption", set by UC-04 when the medical records are complete.
3. The screening rules are configured and active.
4. The applicant has no pending application on the same animal.

Postconditions

1. An application record is saved and linked to the animal, with its screening outcome.
2. If not rejected, the animal is held for 48 hours and the application enters the staff review queue.
3. An entry is written to the access log and a confirmation email is sent to the applicant.

Main Success Scenario

1. Applicant selects an animal from the catalogue (UC-01) and clicks "Apply to Adopt".
2. System checks that the animal is still available and that the applicant has no duplicate application.
3. System displays the adoption questionnaire, asking for housing type, landlord consent, existing pets, hours the animal would be left alone, and prior experience.
4. Applicant fills in the questionnaire and submits it.
5. System validates that all mandatory fields are filled and correctly formatted.
6. System runs UC-06 Screen Questionnaire, which scores the answers against the active rules.
7. The score is above the approval threshold.
8. System saves the application, places a 48 hour hold on the animal and writes a log entry.
9. System sends a confirmation email containing the application reference number.
10. System shows the reference number on screen with a link to book a meeting with the animal.
11. The application is added to the staff review queue for UC-03.
12. Use case ends successfully.

Alternate Flow 7a - Application Rejected by Screening

- 7a1. The score is below the rejection threshold, for example because landlord consent was not given.
- 7a2. System saves the application as Rejected and records which rule failed.
- 7a3. No hold is placed and the animal stays available in the catalogue.
- 7a4. System emails the applicant the outcome, the reason, and guidance on applying again.
- 7a5. System suggests other animals that suit the applicant's answers.
- 7a6. Use case ends. Nothing is added to the staff queue.

Alternate Flow 2a - Animal No Longer Available

- 2a1. At step 2 the system finds the animal has been adopted or sent back to quarantine.
- 2a2. System shows a message saying the animal is no longer available and does not open the questionnaire.
- 2a3. System offers similar animals and returns the applicant to UC-01.
- 2a4. Use case ends without an application being created.